MOHAN RAJ SENTHILNATHAN

PRINCIPAL SOFTWARE ENGINEER
DISTRIBUTED SYSTEMS | AI AGENTS

+91 9043 121 430
Mohansenthil92@gmail.com
Chennai
LinkedIn.com/in/mohanras

PROFESSIONAL OVERVIEW

Principal Software Engineer with 12+ years of experience building scalable, secure distributed systems and cloud-native platforms. Proven expertise in modernizing legacy architecture and developing AI agents using the Claude Agent SDK to automate workflows, reduce manual effort, and improve system efficiency.

WORK EXPERIENCE

Gen (Formerly NortonLifeLock / Symantec) - Chennai **Jan 2017 – Present**

Principal Software Engineer (Jul 2023) | Senior SWE (Aug 2020) | SWE (Jan 2017)

Multi-Tenant Web Portal:

- Architected and delivered a cloud-native multi-tenant platform using .NET Core, React, Azure, Redis, and Cosmos DB; scaled across Norton, Avast, and Avira **serving 400K daily active users.**
- **Built AI-powered test triage** (Claude Agent SDK) and automated defect remediation framework resolving up to 60% of regression failures without human intervention.
- **Built an AI defect-creation tool** (FastAPI, React, AWS Bedrock Claude) that turns free-text bug reports into structured Jira tickets. It uses **Pinecone hybrid search** (dense + sparse) over a curated product-flow knowledge base to identify the right use case, with schema-validated tool calling and Langfuse tracing. Now used to raise 80% of defects.
- Architected a security layer using OpenID Connect and RESTful APIs for distributed multi-tenant auth.
- Optimized Cosmos DB queries (selective fields, custom indexing) and integrated Redis cache for cross-server sync.

Email Guardian:

- Designed Feedback Loop enhancements with automatic re-scans via Pub/Sub; built a high-throughput Metrics API using Kafka and GCP BigQuery for real-time analytics.
- **Built a multi-agent AI service** that detects IMAP settings and produces localized email setup guides on demand. Made email onboarding faster and self-service, lowering support effort and making it easy to scale to new providers.
- Implemented Database Sharding for horizontal scalability, enabling Norton tenant onboarding.
- Migrated Nginx to free tier via custom open source auth modules, reducing licensing costs.
- Developed a microservice orchestrator using RabbitMQ for scalable message-driven workflows, processing ~43M emails scanned per day.

Leadership:

- Mentored a team of 15 engineers, led code reviews, and championed knowledge-sharing sessions and design discussions. Led AI tool adoption including prompt engineering and spec design across the team.

HCL Technologies (Client: Cisco Systems) - Chennai **Mar 2015 – Dec 2016**

Member Technical Staff

- Developed REST APIs and backend services for Cisco Identity Services Engine (ISE).

Amazon Development Center — Chennai **Oct 2013 – Mar 2015**

Device Associate

- Owned module-level testing for Amazon Fire Phone & Kindle devices, automating key test cases to enhance release *quality.*

SKILLS

Languages: C#, Java, Python, JavaScript, Typescript
AI & Agents: Claude Agent SDK, AWS Bedrock, LangGraph, RAG pipelines
Frameworks: .NET Core, Spring Boot, Django, React, Angular, Node.js
Cloud & DevOps: Azure, GCP, AWS, Kubernetes, Docker, Redis, Nginx, TeamCity, Azure DevOps, Git
Data: PostgreSQL, Cosmos DB, Kafka, RabbitMQ
Observability: Prometheus, Grafana, Datadog, Splunk
Architecture: Microservices, Event-Driven, CQRS, Domain-Driven Design, SOLID

ACHIEVEMENTS

Patent
Systems and Methods for Restricting Capture of Self-Portrait Images on Mobile Devices
Patent No: 11528355
Issued: Dec 13. 2022

EDUCATION

B. Tech in Information Technology
Anna University
2013

PROFESSIONAL DEVELOPMENT

Certification
Microsoft Certified: Azure Fundamentals (AZ-900)
2022